

BT4222 Project Interim Report Guideline

August 2026

Due: 11:59 PM, 4 Oct. 2026. Submitted on a group basis via Canvas in .pdf or .docx format.

Purpose

The interim report ensures that your group has properly acquired, explored, and **thoroughly understood the raw datasets**. You must demonstrate that you have actively inspected the data structures, distributions, and impression/interaction dynamics to establish a sound foundation for subsequent feature engineering, baseline benchmarking, and model development. (Note: Baseline benchmarking and model development are not required for the interim report. While you are encouraged to begin conceptualizing feature ideas based on your data exploration, completing full feature engineering is not mandatory at this stage.)

Dataset Requirements

By default, each group can choose one of the following two datasets as introduced in class:

- **JDSearch** (e-commerce search and transaction logs)
- **Microsoft MIND** (news recommendation and impression logs)

Working with either dataset grants access to previous student project source code (e.g., NCF, NMF baselines), which provides your comparative starting point.

Alternative Datasets:

If your group proposes an alternative dataset, it must satisfy the following criteria:

- Explicitly record **impression/exposure logs** (capturing items exposed to the user but not interacted with, in addition to positive interactions).
- Contain a sample size of at least **100,000 records**.
- Still conduct baseline benchmarking and gap diagnosis against past project baselines **conceptually**.

Note on Data Stability:

The dataset and target defined in this report should be the primary foundation for your final project. Conversely, you may implement **minor refinements** regarding your sampling strategy, feature set, or the integration of supplementary data sources. Substantial post-interim changes (e.g., switching datasets or redefining the prediction target due to negligence or misunderstanding) will be subject to mark penalties.

Format of the Project Interim Report

- **Cover Page:** Project Title, Group Number, Full Names of all Group Members.
- **Formatting:** Line spacing: 1.5 lines; Font: Arial, Calibri, or Times New Roman, Size 11; Margins: Normal; Columns: One.
- **Length:** No strict page limit, but avoid unnecessary fluff; keep explanations succinct, fact-based, and evidence-driven.

Information to Be Included in the Interim Report (3 marks)

Part 1. Your Machine Learning Problem

In concise bullet points, formulate your recommendation problem based on the business context and your initial data exploration:

- **Business Objective & ML Alignment:** State the business objective and how it maps to your ML task (e.g., CTR prediction, re-ranking, session conversion).
- **Model Inputs & Target Outputs:** Explicitly define what constitutes an input sample (e.g., user profile, impression slate, session history) and the ground-truth prediction target/label.

Part 2. Datasets & Impression Data Understanding

Prove that your group has thoroughly explored and analyzed the raw dataset. Include the following analyses:

1. Data Overview & Sampling Scheme

- Data source, total sample size, entity counts (unique users, unique items, total sessions/queries), and the time span covered.
- If using a subset, clearly specify your filtering, temporal splitting, or sampling criteria.

2. Impression, Exposure & Interaction Dynamics

To demonstrate a genuine understanding of transaction logs with impressions, provide quantitative distributions or visualizations. For example,

- **Impression Slate & Position Distribution:** e.g., Slate lengths (number of items exposed per impression) and click-through rates across display ranks/positions.
- **Interaction Hierarchy & Funnel:** e.g., Volume and conversion rates across interaction stages (e.g., *impression* → *click* → *detail read* / *cart* → *purchase*).
- **Feedback Semantics:** e.g., How exposed-but-unclicked items (explicit negative signals) are distinguished from unobserved/unexposed items.
- **Sparsity & Temporal Patterns:** e.g., Distribution of user interaction sequence lengths, item popularity distributions (head vs. long-tail), and time gaps between user actions.

3. Feature Dictionary & Descriptive Statistics

Provide structured tables summarizing features you plan to utilize or engineer. For example,

- **Feature Metadata:** Feature name, description, and data type/unit (e.g., seconds, word count, categorical ID).
- **Numeric Features:** Min, max, mean, median, standard deviation, and missing value counts.
- **Categorical Features:** Distinct category counts, representative values, and missing value counts.
- *Note:* If the dataset contains over 50 features, group them by category/domain and report representative features for each category.
- **Text / Corpus / Media (if applicable):** Distribution of sequence/text lengths, vocabulary sizes, or image metadata.